Creating Parse Trees

Constituency-based parsing: how a sentence breaks down into nested phrases.

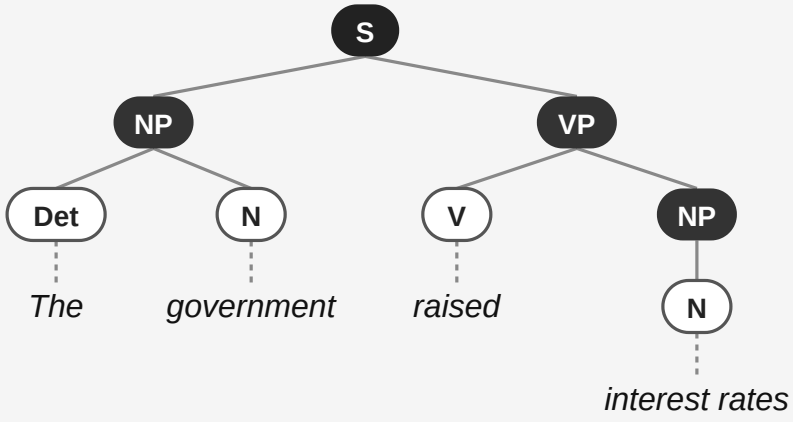
ACTIVITY GUIDANCE

Create a constituency-based parse tree for the following phrases:

1. **The government raised interest rates.**
2. **The internet gives everyone a voice.**
3. **The man saw the dog with the telescope.**

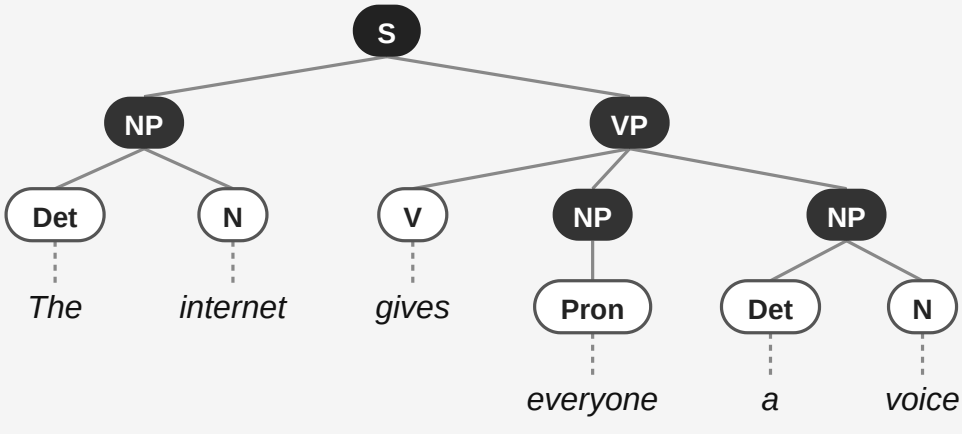
1 The government raised interest rates.

This sentence follows the structure **S → NP + VP**.



2 The internet gives everyone a voice.

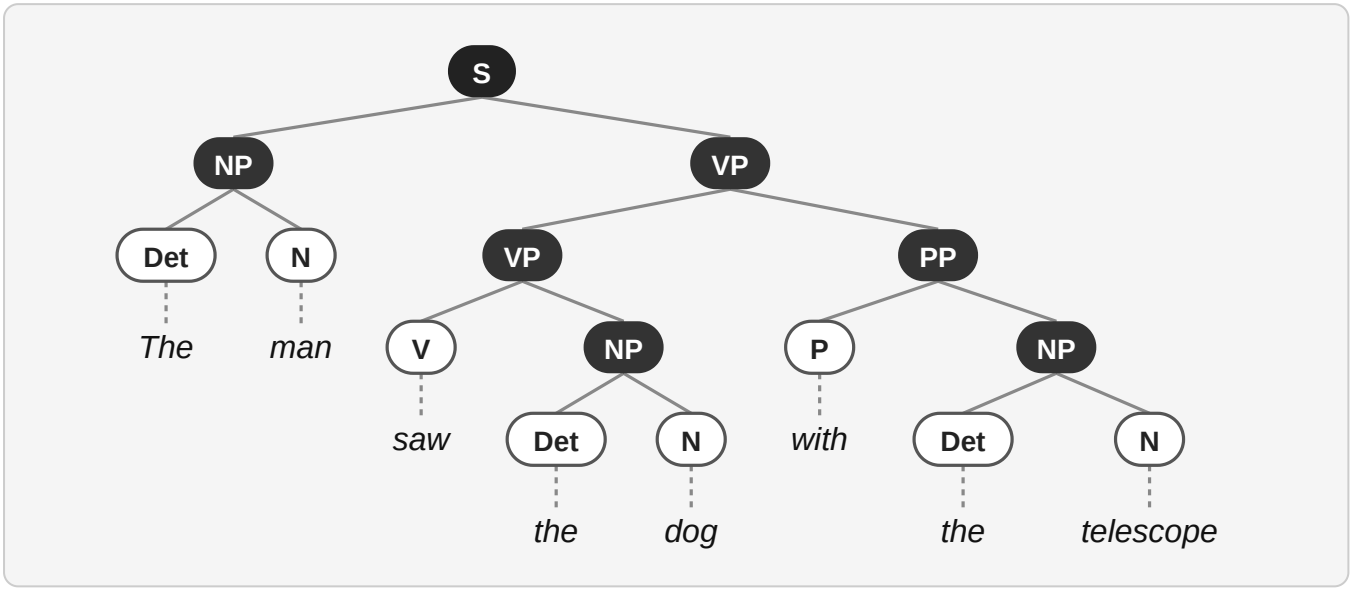
Here, *gives* takes two noun phrases: *everyone* (the indirect object) and *a voice* (the direct object).



3 The man saw the dog with the telescope.

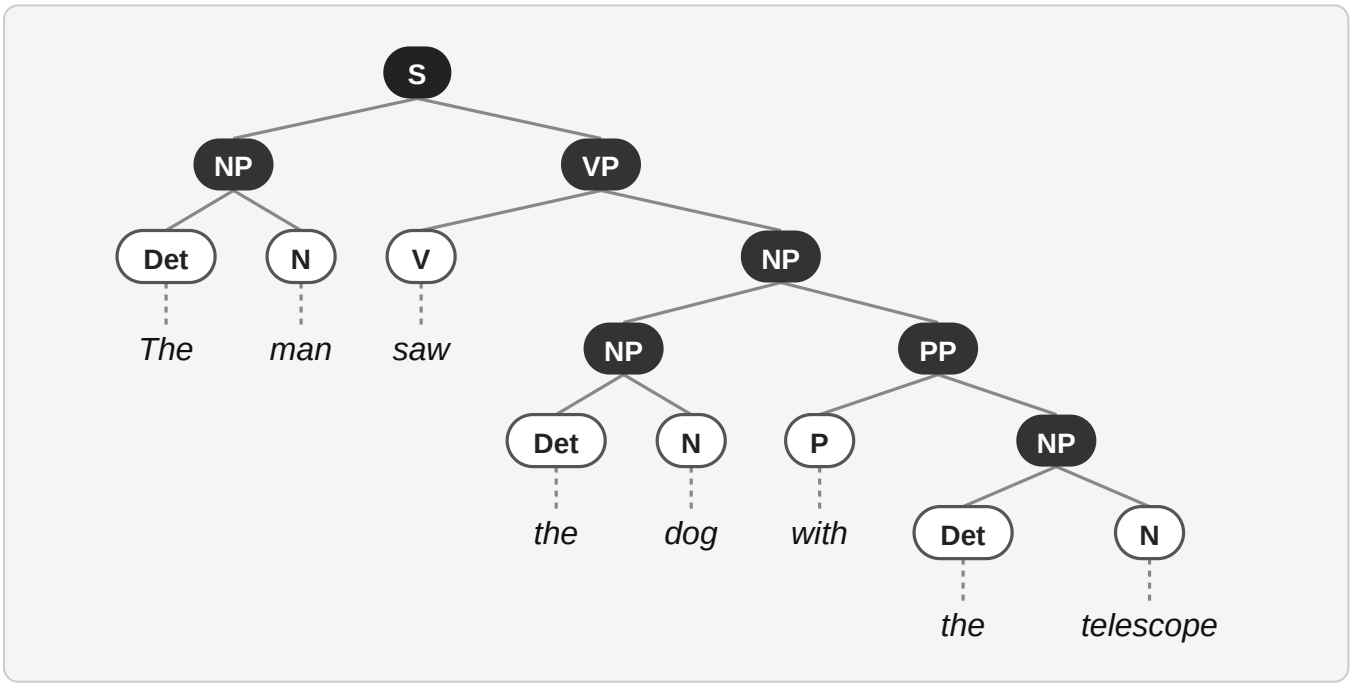
This sentence is structurally ambiguous: *with the telescope* can describe the instrument used to see the dog, or it can describe the dog.

READING A **The man used the telescope.**



The prepositional phrase (PP) attaches to the verb phrase (VP), indicating the instrument used for seeing.

READING B **The dog had the telescope.**



The PP attaches to the noun phrase (NP), specifying which dog the man saw.

E-PORTFOLIO

Key concepts

LABEL	MEANING	EXAMPLE
S	Sentence	*The whole sentence*
NP	Noun phrase	*the government*
VP	Verb phrase	*raised interest rates*
PP	Prepositional phrase	*with the telescope*
Det	Determiner	*the, a*
N	Noun	*government, dog*
V	Verb	*raised, saw*
P	Preposition	*with*
Pron	Pronoun	*everyone*

Same words, two structures

Reading A · PP attaches to VP

The man used the telescope to see the dog.

Reading B · PP attaches to NP

The dog the man saw had the telescope.

CONCLUSION

Constituency-based parsing identifies the hierarchical structure of sentences. The third example demonstrates how different constituent attachments can produce different interpretations of the same sentence.